

ML Project Report: Autism Spectrum Disorder Prediction

1. Introduction

Autism Spectrum Disorder (ASD) is a neurological condition that affects communication, behavior, and social interaction. Early detection is very important for timely treatment and support. Traditional diagnosis methods are slow and require expert analysis. This project uses machine learning to predict ASD quickly and efficiently using screening data.

2. Data Description

The dataset is taken from Kaggle and contains patient screening data.

Features include:

Age, Gender, Ethnicity, Family history, AQ-10 screening scores

Target variable: ASD (Yes/No)

EDA Findings:

- Dataset cleaned and preprocessed
- Categorical data encoded
- AQ score features strongly related to ASD
- Minimal missing values after handling

3. Feature Importance

Using SHAP analysis:

- AQ scores are most important features
- Family history also impacts prediction
- Demographic features have less impact

4. Methodology

Flow of project:



5. Algorithms Used

Logistic Regression – simple and interpretable

KNN – based on nearest neighbors

Naïve Bayes – probability-based

Decision Tree – tree structure model

SVM – finds optimal boundary

Random Forest – ensemble of trees

Gradient Boosting – sequential learning

XGBoost – optimized boosting algorithm

6. Result and Discussion

Evaluation Metrics: Accuracy, Precision, Recall, F1-score

Results (example):

Logistic Regression – 85%

KNN – 82%

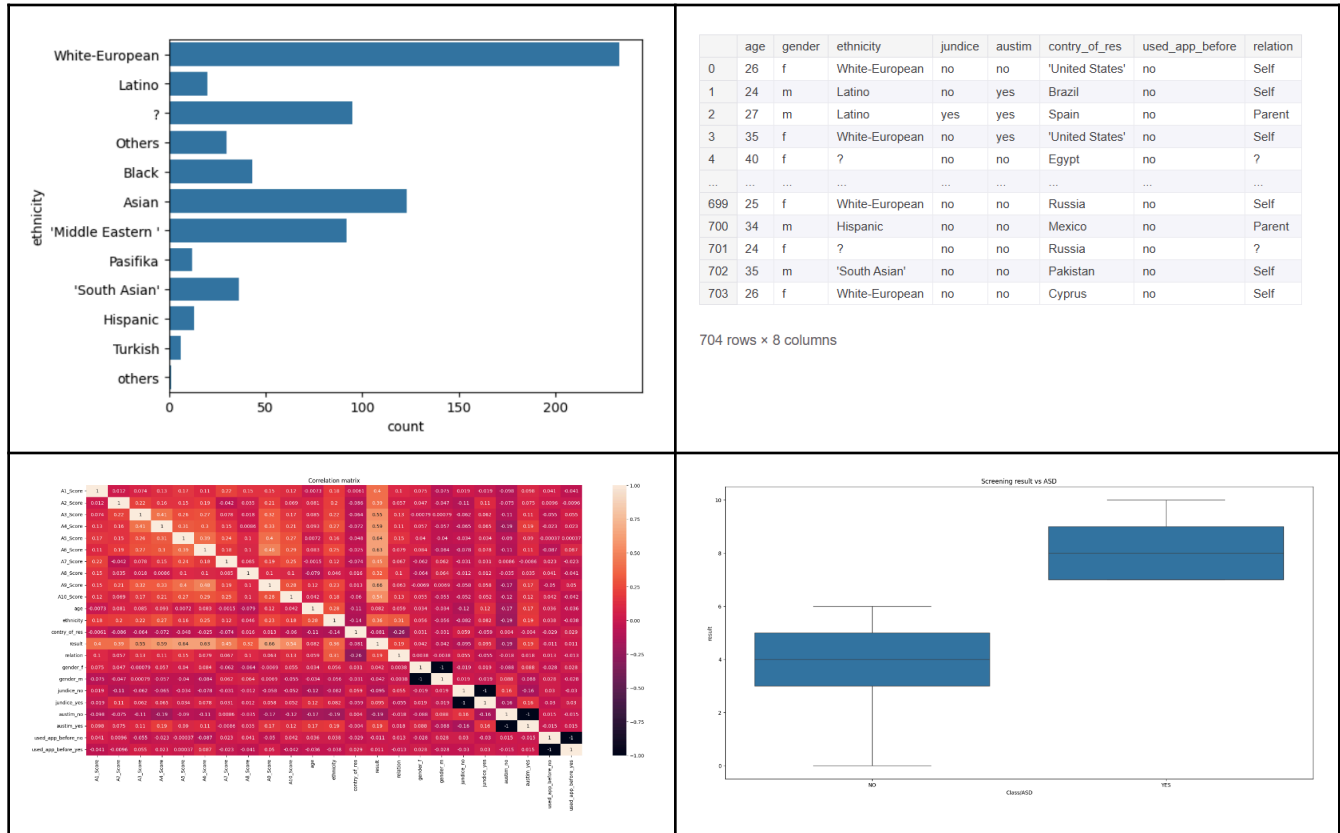
Decision Tree – 88%

Random Forest – 92%

XGBoost – 93%

Conclusion:

Random Forest and XGBoost performed best because ensemble methods reduce overfitting and improve accuracy.



7. Data Source and Tools

Dataset: Kaggle ASD Dataset (<https://www.kaggle.com/code/mrunalijibhakate02/ml-ta-3-4>)

Tools Used:

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, SHAP

AI Tool Used:

ChatGPT